

Machine Learning and Deep Learning - Comprehensive Study Notes

Machine Learning Fundamentals

Machine Learning is a branch of Artificial Intelligence that enables computers to learn patterns from data without being explicitly programmed. The primary objective of machine learning is to build systems capable of generalizing from historical observations to unseen situations. A complete machine learning pipeline includes data collection, preprocessing, exploratory data analysis, feature engineering, model training, evaluation, deployment, and monitoring. Supervised learning uses labeled datasets to learn relationships between input features and target outputs. Popular algorithms include Linear Regression, Logistic Regression, Decision Trees, Random Forests, XGBoost, LightGBM, CatBoost, Support Vector Machines, and Neural Networks. Unsupervised learning discovers hidden patterns in data through clustering, dimensionality reduction, and anomaly detection. Common techniques include K-Means, DBSCAN, PCA, and Autoencoders. Feature engineering remains one of the most important aspects of machine learning. Transforming raw data into meaningful representations can improve performance significantly. Scaling, normalization, handling missing values, encoding categorical variables, and feature selection are critical steps. Machine Learning is a branch of Artificial Intelligence that enables computers to learn patterns from data without being explicitly programmed. The primary objective of machine learning is to build systems capable of generalizing from historical observations to unseen situations. A complete machine learning pipeline includes data collection, preprocessing, exploratory data analysis, feature engineering, model training, evaluation, deployment, and monitoring. Supervised learning uses labeled datasets to learn relationships between input features and target outputs. Popular algorithms include Linear Regression, Logistic Regression, Decision Trees, Random Forests, XGBoost, LightGBM, CatBoost, Support Vector Machines, and Neural Networks. Unsupervised learning discovers hidden patterns in data through clustering, dimensionality reduction, and anomaly detection. Common techniques include K-Means, DBSCAN, PCA, and Autoencoders. Feature engineering remains one of the most important aspects of machine learning. Transforming raw data into meaningful representations can improve performance significantly. Scaling, normalization, handling missing values, encoding categorical variables, and feature selection are critical steps. Machine Learning is a branch of Artificial Intelligence that enables computers to learn patterns from data without being explicitly programmed. The primary objective of machine learning is to build systems capable of generalizing from historical observations to unseen situations. A complete machine learning pipeline includes data collection, preprocessing, exploratory data analysis, feature engineering, model training, evaluation, deployment, and monitoring. Supervised learning uses labeled datasets to learn relationships between input features and target outputs. Popular algorithms include Linear Regression, Logistic Regression, Decision Trees, Random Forests, XGBoost, LightGBM, CatBoost, Support Vector Machines, and Neural Networks. Unsupervised learning discovers hidden patterns in data through clustering, dimensionality reduction, and anomaly detection. Common techniques include K-Means, DBSCAN, PCA, and Autoencoders. Feature engineering remains one of the most important aspects of machine learning. Transforming raw data into meaningful representations can improve performance significantly. Scaling, normalization, handling missing values, encoding categorical variables, and feature selection are critical steps.



Deep Learning and Neural Networks

Deep Learning is a subset of machine learning that uses multi-layer neural networks. Neural networks are inspired by biological neurons and consist of interconnected computational units. Each neuron computes a weighted sum of inputs and applies an activation function. Common activation functions include ReLU, Sigmoid, Tanh, GELU, and Swish. Training occurs through forward propagation and backpropagation. The forward pass computes predictions while backpropagation calculates gradients. Gradient Descent and its variants such as SGD, Adam, AdamW, RMSProp, and Adagrad update model parameters. Deep learning has transformed computer vision, natural language processing, speech recognition, medical imaging, robotics, recommendation systems, and generative AI. Modern models often contain millions or billions of parameters.

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Python Example: Linear Regression

```
from sklearn.linear_model import LinearRegression

model = LinearRegression()
model.fit(X_train, y_train)

predictions = model.predict(X_test)

print(predictions[:5])
```

The above code trains a Linear Regression model using scikit-learn. The `fit()` method learns model coefficients from training data. The `predict()` method generates predictions on unseen samples. Regression models are commonly used in price prediction, forecasting, risk analysis, and business intelligence systems. Model evaluation is performed using metrics such as MAE, MSE, RMSE, and R^2 . Cross-validation improves robustness by evaluating models across multiple data splits. The above code trains a Linear Regression model using scikit-learn. The `fit()` method learns model coefficients from training data. The `predict()` method generates predictions on unseen samples. Regression models are commonly used in price prediction, forecasting, risk analysis, and business intelligence systems. Model evaluation is performed using metrics such as MAE, MSE, RMSE, and R^2 . Cross-validation improves robustness by evaluating models across multiple data splits. The above code trains a Linear Regression model using scikit-learn. The `fit()` method learns model coefficients from training data. The `predict()` method generates predictions on unseen samples. Regression models are commonly used in price prediction, forecasting, risk analysis, and business intelligence systems. Model evaluation is performed using metrics such as MAE, MSE, RMSE, and R^2 . Cross-validation improves robustness by evaluating models across multiple data splits. The above code trains a Linear Regression model using scikit-learn. The `fit()` method learns model coefficients from training data. The `predict()` method generates predictions on unseen samples. Regression models are commonly used in price prediction, forecasting, risk analysis, and business intelligence systems. Model evaluation is performed using metrics such as MAE, MSE, RMSE, and R^2 . Cross-validation improves robustness by evaluating models across multiple data splits.

CNNs, RNNs, Transformers and Modern AI

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sequences in parallel and capture long-range relationships efficiently. Modern Large Language Models such as GPT, Gemini, Claude, LLaMA, and Mistral are transformer-based systems. Retrieval-Augmented Generation combines vector search with language models to improve factual accuracy. Enterprise AI systems increasingly use Graph RAG, Agentic AI, Multi-Agent Systems, Knowledge Graphs, and Self-Correcting RAG pipelines. Future research directions include multimodal reasoning, autonomous agents, edge AI, federated learning, and trustworthy AI.